Bayesian Network Analysis for Loss of Controlled Flight (LCF)

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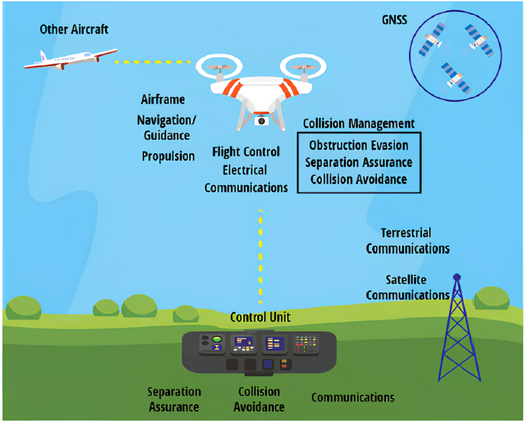
1. System Overview

- Bayesian Network (BN) developed to estimate the probability of **Loss of Controlled Flight (LCF)** in a UAS.
- Structure derived from a Fault Tree in the reference article.
- Includes software faults, environmental hazards, communication issues, and human errors.



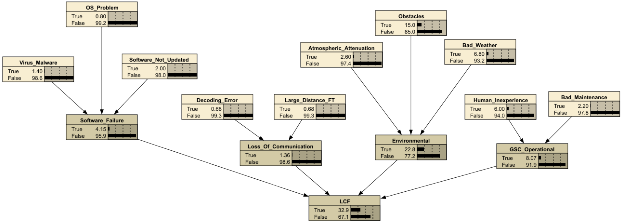
2. Model Structure

- **Basic events**: OSProblem, VirusMalware, Obstacles, BadWeather, HumanInexperience, etc.
- **Intermediate nodes**:
  - SoftwareFailure
  - LossOfCommunication
  - Environmental
  - GSCOperational
- **Top event**: LCF
- All CPTs modelled using **Noisy-OR**.
- Prior probabilities extracted from literature.

- Computed unconditional probability:

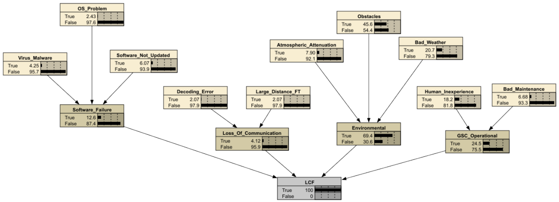
$$P(\mathrm{LCF} = \texttt{True}) \approx 0.329$$

- Environmental hazards are the dominant contributors.
- Provides baseline for posterior, sensitivity, and reliability analyses.



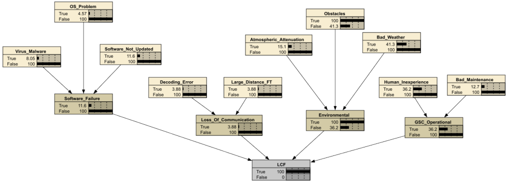
4. A Posteriori Analysis

- Conditioning on **LCF** = **True**.
- Posterior probabilities of intermediate nodes:
  - Environmental: 0.694
  - GSCOperational: 0.245
  - SoftwareFailure: 0.126
  - LossOfCommunication: 0.041
- Environmental hazards and operator errors emerge as main contributors.



5. Most Probable Explanation (MPE)

- MPE computed under evidence **LCF = True**.
- Most likely failed nodes:
  - Obstacles ≈ 1.00
  - Environmental ≈ 1.00
  - BadWeather ≈ 0.413
  - HumanInexperience ≈ 0.362
  - GSCOperational ≈ 0.362
- Scenario dominated by environmental hazards + operator inexperience.



6. Reliability and Conclusions

- Failure rate:

$$\lambda \approx 0.399 \text{ failures/mission}$$

- MTBF:

$$\text{MTBF} \approx 2.5 \text{ missions}$$

- Key insights:
  - Environmental hazards are the primary drivers of LCF.
  - Human factors also significantly contribute.
  - Improvements needed: sensing, obstacle detection, operator training.